

Apriori Algorithm based Association Rule Mining to Enhance Small-Scale Retailer Sales

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Abstract— This research aims to assist small online gift store retailers who have not yet adopted a recommender system by leveraging association rules to offer personalized product recommendations to their customers. Using the Apriori algorithm and association rule mining techniques in Python, a dataset of UK online gift store retailer sales transactions was analyzed to identify frequent itemsets and association rules. The main objective of this research is to enable retailers to better manage their inventory and boost sales by offering products that customers are likely to purchase based on their previous buying behaviour. The result presents the top 10 frequent itemsets and their support values, along with a list of association rules. The result indicates that candlelight holders, bags, and tableware items are frequently purchased together. This research provides valuable insights for small online gift store retailers to enhance their sales performance and customer satisfaction by leveraging association rules to offer personalized product recommendations.

Keywords— *Apriori Algorithm, Association Rule, Confidence, Decision making, Recommender System*

I. INTRODUCTION

In the face of declining sales in the gift retail industry, it has become increasingly important for retailers to adopt data-driven approaches to help them better manage their products and increase their sales opportunities. With the rise of online shopping, traditional methods of product management and recommendation have become inadequate, making it crucial for retailers to leverage digital footprints left by customers as they browse, click, and purchase products.

Works accomplished in [1], [2] and [3] discuss the declining sales of the gift retail industry and explore the factors contributing to this trend. They all highlight the impact of changing consumer behaviour, such as the rise of online shopping, as well as market forces, such as economic conditions and globalization. Additionally, the papers offer recommendations for retailers to adapt to the changing market and remain competitive, such as focusing on customer experience, providing unique products and services, and diversifying their offerings.

The sale of one product can influence the sale of other related products, making it crucial for retailers to identify these associations and leverage them to maximize sales opportunities. For instance, if a customer purchases a mobile phone, it is highly likely that the customer will also purchase other accessories such as a cover, screen protector, or

headphones. By analyzing these associations, retailers can optimize their product offerings and provide more personalized product recommendations to customers, resulting in increased sales. However, manually identifying the associations between gift products is a daunting task for retailers.

To address these challenges, recommendation systems have become increasingly popular in the online sales domain. These systems help consumers make informed purchase decisions by recommending products based on their browsing or purchase history. Traditionally, recommender systems relied on word-of-mouth, reviews, surveys, and other reading materials. However, as more consumers move to online platforms as their preferred method of shopping, these systems now leverage digital footprints left by customers as they browse, click, and purchase products. Retailers can easily gather a trail of data through customer purchases or browsing information, which can be used to extract hidden patterns and insights that can help them better understand their customers' behaviour. By leveraging customer data and identifying product associations, online retailers can provide more personalized product recommendations, improve customer satisfaction, and ultimately increase sales.

This research aims at contributing some insights for small online gift store retailers that have yet to adopt a recommender system by leveraging association rules and data mining techniques. Valuable insights about customer behaviour and preferences can be obtained and used to optimize inventory management and boost sales. As online shopping continues to grow in popularity, it is becoming increasingly important for retailers to offer personalized product recommendations to their customers. This research involves analysis of the purchasing patterns of gift retail customers. The combination of items that are frequently purchased is studied. The key aspects necessary for a product recommendation for retailers are identified.

II. LITERATURE REVIEW

Research in [4] studies the sales data from grocery stores in Turkey and the data was analyzed using the Apriori algorithm. Their findings indicated that 77% of cucumber areas receive tomatoes, and Purslane and 55% of tomato areas receive parsley. This illustrates how the algorithm was used to identify patterns and associations between different products in the dataset.

Similarly, researchers in [5] conducted a study using the FP growth algorithm and Apriori algorithm to analyze a dataset from a grocery retail store in Mumbai. They found that the Apriori algorithm outperformed the FP growth algorithm in terms of processing time while producing similar results. This suggests that the Apriori algorithm may be a more efficient method for analyzing large datasets in the grocery retail industry.

Researchers in [2] conducted a case study on a computer retail store in Indonesia using the Apriori algorithm. The study was performed using the PHP programming language, and the dataset consisted of three months' worth of sales data from January to March 2019. The aim of the study was to identify patterns and associations between different products sold in the store.

The findings of the case study revealed that there were two-item combinations with a high confidence value of 100% and a lift ratio value of 3.39. Specifically, customers who purchased a Laptop Charger were likely to also purchase a keyboard. For three-item combinations, customers who purchased Joystick and Laptop were likely to also purchase a mouse with a high confidence value of 100% and a lift ratio value of 2.17. These findings suggest that certain products are often purchased together, which can provide valuable insights into customer behaviour and preferences.

A study [6] proposed an improved version of the Apriori algorithm and tested it on a dataset from the Taobao e-commerce clothing store. The dataset contained 233 sales transactions with 68 different products. The aim of the study was to improve the efficiency and accuracy of the Apriori algorithm by reducing the number of irrelevant items in the dataset. The findings of the study showed that the improved Apriori algorithm led to time savings of approximately 20% compared to the original algorithm, with an increase in accuracy of around 10%. These results suggest that the improved algorithm is a more efficient and accurate method for analyzing sales data in e-commerce clothing stores.

In the study [7] there was proposed a study of the supervised and the unsupervised machine learning algorithms. The high rate of applicability of the Apriori Algorithm for a wide array of problems has been elaborated. The research explains the comparative success rate of the algorithm against several algorithms used in the field.

Another study, [8] discussed the use of the Apriori algorithm in stock handling. The study proposed that association rules with a minimum support value of 50% and a minimum confidence of 60% could be used as a reference in determining product combinations for store shelves arrangement. The findings of the study suggest that the Apriori algorithm can be an effective tool for optimizing store shelf arrangement and improving the efficiency of stock handling in retail businesses.

In a recent study by [9], the authors compared the performance of Apriori, FP-Growth, and two newer algorithms, Eclat and PrefixSpan, on different datasets. The results showed that Apriori and FP-Growth outperformed Eclat and PrefixSpan in terms of execution time and memory usage,

particularly for smaller datasets. The authors also noted that Apriori was more resource-efficient than the newer algorithms, as it generated fewer candidate itemsets.

Similarly, a study by [10] compared the performance of several frequent itemset mining algorithms, including Apriori and FP-Growth, on various datasets. The results showed that Apriori was the most efficient algorithm for mining frequent itemsets in small datasets, while FP-Growth performed better for larger datasets. The use of Apriori Algorithms for definitive problem solving is well established in the works of [11] and [13]. Work done in [11] demonstrates the application of the algorithm for a recommendation system has been developed for depression consultancy. Similar work done in [12] established the use of the algorithm to propose recommendations for the architectural design and development. Work done in [13] establishes the application of the machine learning strategy for real time problem solving such as pot-hole detection.

III. SYSTEM ARCHITECTURE AND METHODS

In this research, a case study methodology was adopted. The obtained data used in this study is a secondary dataset from Kaggle.com. Data pre-processing will be conducted, followed by exploratory data analysis (EDA). This research will also use Python programs to perform the analysis. The Apriori algorithm is then used to generate association rules that identify which products are frequently purchased together. These association rules will be studied to generate personalized product recommendations for customers based on their past purchasing behaviour.

A. Data Pre-Processing

The dataset is taken from a UK online gift retailing store, containing 541909 transactions over the period of one year. Fig. 1 shows a subset of the data for the structure of the dataset as processed in Python. Table I shows the features. There are 8 attributes namely, InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID and Country.

RangeIndex: 541909 entries, 0 to 541908				
Data columns (total 8 columns):				
#	Column	Non-Null Count		Dtype
0	InvoiceNo	541909 non-null		object
1	StockCode	541909 non-null		object
2	Description	540455 non-null		object
3	Quantity	541909 non-null		int64
4	InvoiceDate	541909 non-null		object
5	UnitPrice	541909 non-null		float64
6	CustomerID	406829 non-null		float64
7	Country	541909 non-null		object

Fig. 1. Structure of dataset

TABLE I. FEATURES OF DATASET

Attributes	Description
InvoiceNo	A 6-digit unique transaction ID. The order is considered cancelled if starts with 'C'
StockCode	A 5-digit unique number assigned to individual product
Description	Description of the product
Quantity	Number of products sold in each transaction
InvoiceDate	Date and time of the invoice issued
UnitPrice	Price of one unit of the product
CustomerID	5-digit unique customer ID assigned to individual customer
Country	Country of residence of the customer

Upon examining for missing values, it is observed that there are 1454 transactions with null values for the 'Description' attribute and 135080 null values for the 'Customer ID' attribute. As the dataset is large enough for analysis purposes, null values will be dropped from the dataset. Similarly, for transactions that are cancelled by customers, transactions with zero value 'UnitPrice', these transactions will be dropped from the dataset as well as these transactions will not serve any purpose for the analysis. The new variable 'Sales' is created by multiplying the unit price and quantity of the product. Another step that was done was to drop all transactions that have only one product sale as the main study of this paper is to analyze products that are bought together. Finally, the remaining dataset consists of 159940 transactions.

B. Exploratory Data Analysis (EDA)

Following pre-processing, EDA is performed to further understand the characteristics and main features of the data. Several insights are observed. In Fig. 2, it is observed that sales start to spike from the month of November. This may be due to seasonal reasons as Christmas month is the month of gifting.

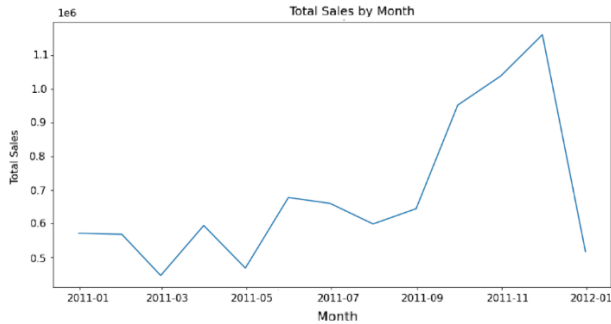


Fig. 2. Line chart showing total sales by month

C. Association Rule Mining

Numerous works have been conducted on the subject of recommender systems, with collaborative filtering, association-ruled recommendation, and content-based recommendation being some of the most commonly used algorithms.

Collaborative filtering is one of the earliest recommendation algorithms, with Tapestry being one of the earliest systems that utilized this approach to filter e-mails (Goldberg et al., 1992). Collaborative filtering focuses on making recommendations based on preferences shared by others. This method has proven to be successful in the market, with many large businesses such as Amazon, YouTube, and Netflix adopting it (Linden et al., 2003).

The association rule is often denoted in the expression form of $X \rightarrow Y$ such that X and Y represent two sets of products (Sarwar et al., 2000). There are a few evaluation metrics involved as follows

- **Support** - It is a measure of the percentage of how frequently the combination items occur. This can be obtained by using the following formula (1)

$$\text{Support} = \frac{\text{number of transactions containing } X \text{ and } Y}{\text{total transactions}} \quad (1)$$

- **Confidence** - It is a measure of the percentage of the relationship strength between the combination items. This can be obtained by using the following formula (2)

$$\text{Confidence} = \frac{\text{number of transactions containing } X \text{ and } Y}{\text{total transactions containing } X} \quad (2)$$

Examples of association rules:

Phone $\Rightarrow$ Casing [support = 2%, confidence = 70%]

D. Apriori Algorithm

One of the simple methods used for association rule mining is the Apriori algorithm that was introduced by Agrawal, R. and Srikant, R. (1994). It generates candidate item sets by successively extending shorter item sets and uses a scan test methodology to identify frequent item sets. Each size of the candidate item set requires a scan to check if it matches the minimum support threshold. The algorithm has two steps, as described by Guo et al. (2017).

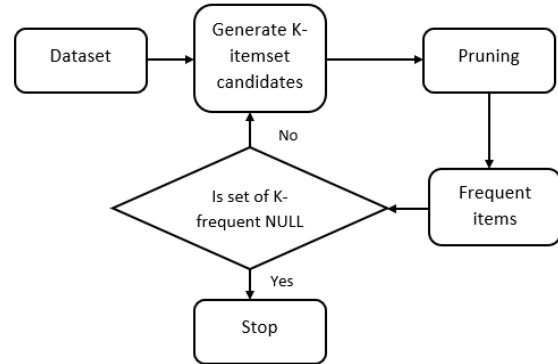


Fig. 3. Steps in Apriori Algorithm

IV. RESULTS

In order to identify products that are frequently sold together, the Apriori algorithm was performed on the post-processed dataset with a 0.03 minimum support value. A minimum support of 0.03 means that only item sets that occur in at least 3% of the transactions will be selected. The resulting top 10 item sets ranked by support are presented in Fig. 4. The itemset with the highest support of 0.123573 was "WHITE HANGING HEART T-LIGHT HOLDER", which means that it appears in approximately 12.36% of all transactions in the dataset. Other highly supported item sets included "JUMBO BAG RED RETROSPOT," "REGENCY CAKESTAND 3

TIER," "PARTY BUNTING," and "ASSORTED COLOUR BIRD ORNAMENT" shown in Fig. 5.

```
>>> frequent_itemsets.head(10)
      support  itemsets
101  0.123573  (WHITE HANGING HEART T-LIGHT HOLDER)
44   0.095041  (JUMBO BAG RED RETROSPOT)
82   0.091916  (REGENCY CAKESTAND 3 TIER)
72   0.085598  (PARTY BUNTING)
6    0.085394  (ASSORTED COLOUR BIRD ORNAMENT)
58   0.073845  (LUNCH BAG RED RETROSPOT)
88   0.066236  (SET OF 3 CAKE TINS PANTRY DESIGN )
52   0.065421  (LUNCH BAG BLACK SKULL.)
64   0.061549  (NATURAL SLATE HEART CHALKBOARD )
29   0.061141  (HEART OF WICKER SMALL)
```

Fig. 4. Top ten frequent itemsets

The association rule mining was performed on the retail dataset, resulting in a set of rules with a minimum threshold of

1 for lift metric. Lift is a measure of the strength of association between two items in a rule. It compares the probability of the presence of both the antecedent and the consequent in a transaction to what would be expected if the two items were independent. A lift value greater than 1 indicates that the presence of the antecedent in a transaction increases the likelihood of the presence of the consequent. The result in Fig. 6 showed the relationship rule with the antecedent (ROSES REGENCY TEACUP AND SAUCER) and consequent (GREEN REGENCY TEACUP AND SAUCER) have the highest lift of 17.3981 indicating that the two itemsets are highly correlated and likely to be purchased together. Another interesting rule was found between (LUNCH BAG PINK POLKADOT) and (LUNCH BAG RED RETROSPOT), with a lift value of 7.5034, suggesting that customers who purchased one of the lunch bags are highly likely to purchase the other.

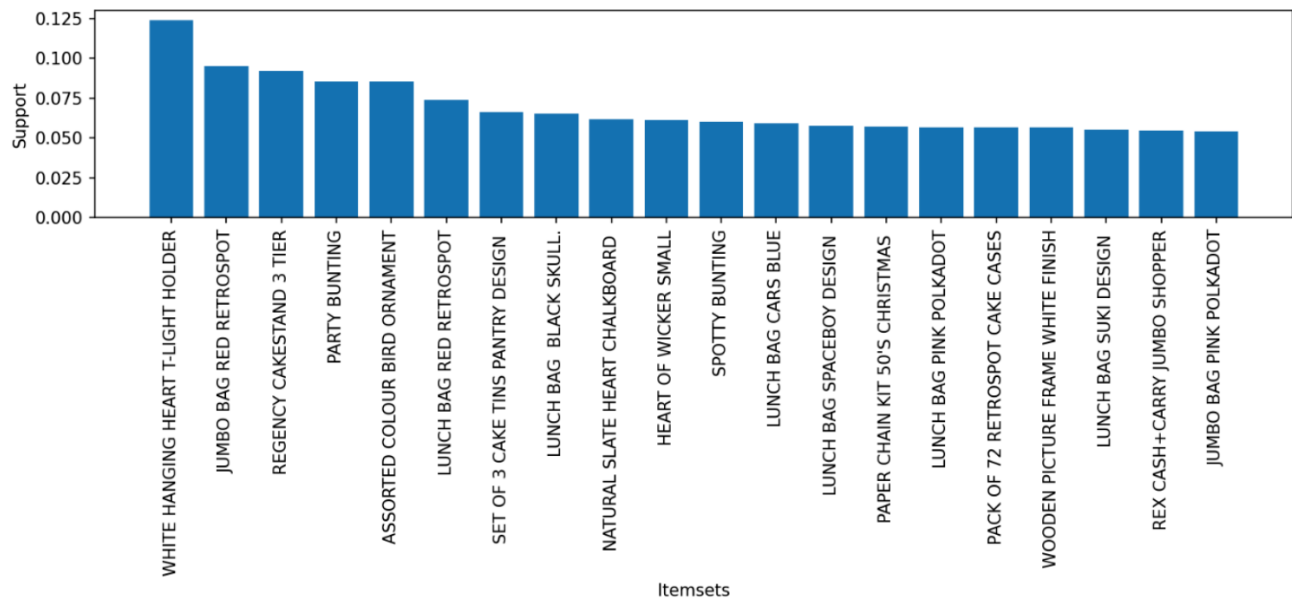


Fig. 5. Bar chart of itemsets and support

	antecedents	consequents	support	confidence	lift	conviction
0	(ROSES REGENCY TEACUP AND SAUCER)	(GREEN REGENCY TEACUP AND SAUCER)	0.031590	0.705615	17.398068	3.259139
1	(GREEN REGENCY TEACUP AND SAUCER)	(ROSES REGENCY TEACUP AND SAUCER)	0.031590	0.778894	17.398068	4.320249
2	(LUNCH BAG PINK POLKADOT)	(LUNCH BAG RED RETROSPOT)	0.031318	0.554087	7.503361	2.076983
3	(LUNCH BAG RED RETROSPOT)	(LUNCH BAG PINK POLKADOT)	0.031318	0.424103	7.503361	1.638276
4	(JUMBO BAG PINK POLKADOT)	(JUMBO BAG RED RETROSPOT)	0.033899	0.626884	6.595953	2.425413
5	(JUMBO BAG RED RETROSPOT)	(JUMBO BAG PINK POLKADOT)	0.033899	0.356683	6.595953	1.470386
6	(LUNCH BAG RED RETROSPOT)	(LUNCH BAG BLACK SKULL.)	0.031793	0.430543	6.581090	1.641175
7	(LUNCH BAG BLACK SKULL.)	(LUNCH BAG RED RETROSPOT)	0.031793	0.485981	6.581090	1.801792

Fig. 6. Result of Association Rules

V. DISCUSSION AND CONCLUSION

The result of this research presents the top 10 frequent itemsets along with their support values and a list of association rules of the studied UK online gift retailer sales transactions. The frequent itemsets are obtained from the Apriori algorithm. The support value indicates the proportion of transactions in which the itemset appears. From the top 10 frequently bought itemsets, it can be observed that candle light holder has the highest support value followed by bag and tableware items. This indicates that these items are frequently purchased together. Among the association rules, it can be observed that the rule with the highest lift value is the one between "ROSES REGENCY TEACUP AND SAUCER" and "GREEN REGENCY TEACUP AND SAUCER". The lift value of 17.39 indicates a strong association between these two items. Similarly, the rule between "JUMBO BAG PINK POLKADOT" and "JUMBO BAG RED RETROSPOT" has a

high lift value of 6.59, indicating a strong association between these items.

This analysis will enable gift store retailers to gain insights into which products are most commonly purchased in the UK region and when these purchases typically occur. This information can then be used to optimize product placement and promotions to increase sales and profitability. The outcome of the research findings will provide valuable insights into customer behaviour and product preferences that can be used to optimize inventory management and product recommendations. The practical implications of the research include increased sales and profitability for small online gift store retailers, as well as improved customer satisfaction and loyalty.

In conclusion, the Apriori algorithm has been used to obtain frequent itemsets and association rules from the given transaction data. The results indicate strong associations between certain items, which can be leveraged by gift store retailers to increase sales and improve customer experience. By identifying these patterns, gift store retailers can offer personalized product recommendations to their customers, thus increasing the likelihood of intra-selling and up-selling. This analysis will also enable retailers to optimize their inventory management by ensuring that commonly purchased products are stocked together. By providing personalized product recommendations, retailers can increase customer loyalty and profitability.

There are a few limitations in this research as the dataset involved consists of transactions that happened in the span of one year. This may not fully capture long-term trends or patterns. A higher sample size may help to improve the accuracy and reliability of the analysis. Also, the dataset has limited variables which do not include demographic data. This is an important variable that can be considered to incorporate for further study as it can provide more insights into the factors that affect sales and help understand customer behaviour.

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